

Project Design Phase
Solution Architecture

Date	11 July 2026
SID	SWUID2026230011
Project Name	Customer Care Registry (MERN Stack)
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. For the Customer Care Registry, it defines how customers, support agents, and administrators interact with a MERN stack application to raise, track, resolve, and analyze support tickets. Its goals are to:

- Find the best tech stack (MongoDB, Express.js, React.js, Node.js) to solve the problem of scattered, untracked customer support.
- Describe the structure, behavior, and interaction of the frontend, backend, database, and third-party services to project stakeholders.
- Define features, development phases (auth, ticketing, notifications, analytics), and solution requirements.
- Provide specifications according to which the solution is built, deployed, and maintained.

Solution Architecture Diagram — Customer Care Registry:

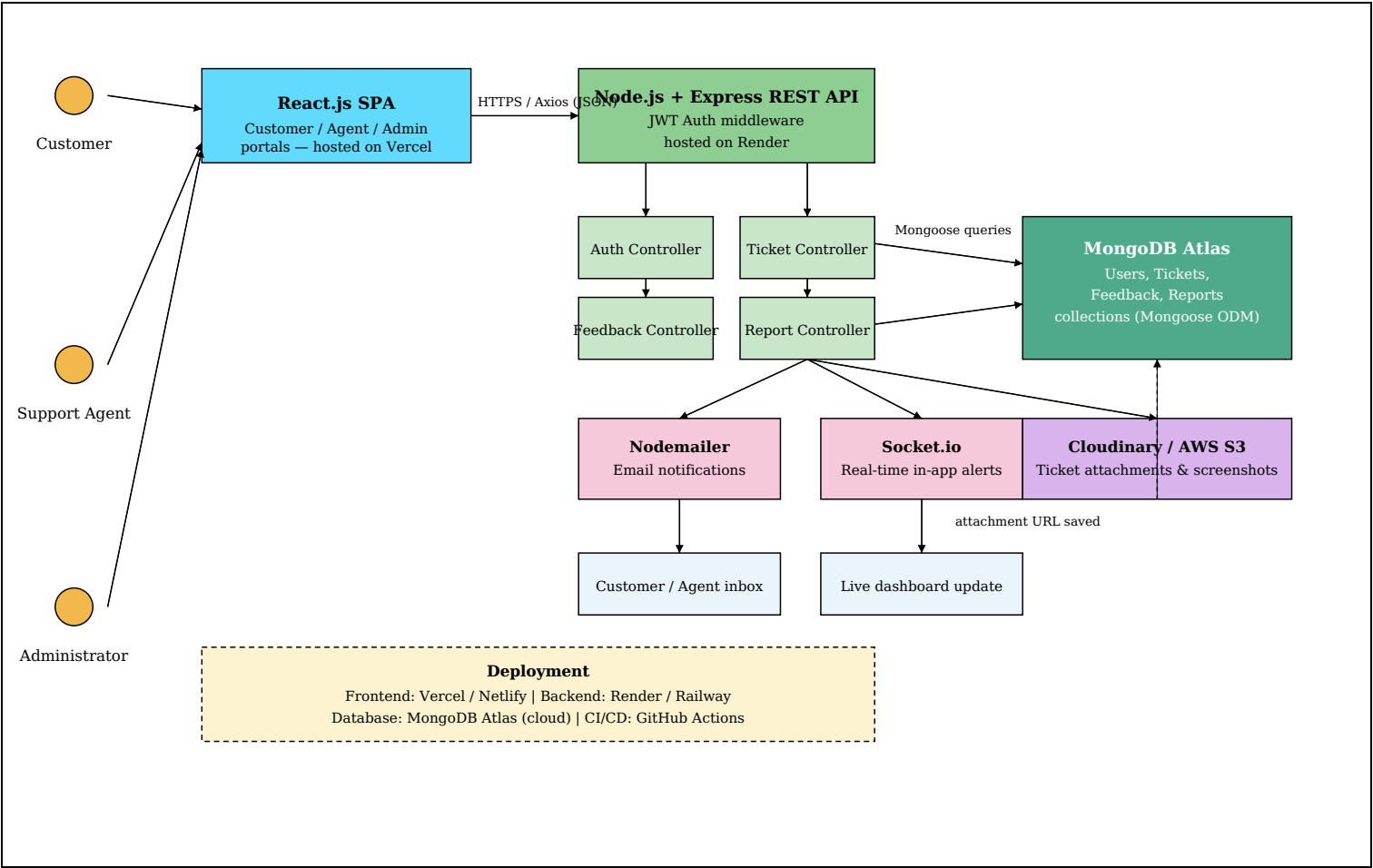


Figure 1: Architecture and data flow of the Customer Care Registry (MERN stack) application.

Flow summary: Customers, agents, and admins interact through role-based React.js portals. All requests go over HTTPS to a Node.js/Express REST API secured with JWT. Controllers handle authentication, ticket management, feedback, and reporting, persisting data in MongoDB Atlas via Mongoose. Ticket updates trigger email notifications (Nodemailer) and real-time dashboard updates (Socket.io); attachments are stored in Cloudinary/S3 with only the URL saved in MongoDB. The frontend is deployed on Vercel/Netlify and the backend on Render/Railway, with MongoDB Atlas as the managed cloud database and GitHub Actions handling CI/CD.

Reference: [MERN stack & REST API architecture patterns](#) (adapted for Customer Care Registry)